

MATH120-24S2 Discrete Mathematics

Assignment 11

Due: 11:59pm Friday 18 October

Submit a single PDF document (1 file only) to the submission link in Learn. It is your responsibility to ensure that your submission is clear and easily readable, and that you upload the correct file by the due date.

1. You are playing a game of poker with a standard deck of 52 cards, with four suits and 13 denominations. A poker hand is a set of 5 cards, their order doesn't matter.
 - (a) A flush* is a poker hand that consists of five cards of the same suit. For example, $\{2\spadesuit, 5\spadesuit, 8\spadesuit, 9\spadesuit, Q\spadesuit\}$ is a flush. What is the probability that a randomly selected poker hand is a flush?
*include flush hands with five-in-a-row in your calculations.
 - (b) What is the probability that a randomly selected poker hand does not contain a 7?

Solution:

Recall that there are a total of $\binom{52}{5}$ possible poker hands.

- (a) There are 4 ways to choose the suit for the flush, and $\binom{13}{5}$ ways to choose the five values that appear. So the probability of randomly selecting a flush is

$$\frac{4\binom{13}{5}}{\binom{52}{5}} = \frac{33}{16660}.$$

- (b) Avoiding the four 7's leaves 48 cards to make a poker hand from with no 7. There are $\binom{48}{5}$ ways to do this, and this makes the probability that we are looking for

$$\frac{\binom{48}{5}}{\binom{52}{5}}.$$

2. There is a building with 4 floors, on each floor there are 5 apartments and in each apartment lives 1 person. Given that 4 people living in the building meet in the entrance, we want to know the probability that they all live on different floors.

- (a) ChatGPT gives the following response. Where is the mistake? Think about ordered/unordered sampling.

Response:

Step 1: Total number of people

The building has 4 floors, each with 5 apartments, and each apartment has 1 person living in it. Therefore, the total number of people living in the building is:

Total people = 4 floors $\times$ 5 apartments per floor = 20 people.

Step 2: Picking 4 people

We are considering a group of 4 people who meet at the entrance. The total number of ways to select 4 people out of 20 is: Total combinations = $\binom{20}{4}$.

Step 3: Calculating the favorable outcomes We want the 4 people to live on different floors. Here's the reasoning:

- The first person can live on any floor.
- The second person must live on a different floor. There are 3 floors left, and 5 apartments per floor, so there are 15 possible choices.
- The third person must also live on a different floor. Now, there are 2 floors left, and 5 apartments per floor, so 10 choices.
- The fourth person must live on the remaining floor. That gives 5 choices.

The number of favorable combinations is:

Favorable combinations = $20 \times 15 \times 10 \times 5$.

Step 4: Probability

The probability that all 4 people live on different floors is the ratio of favorable outcomes to the total combinations: Probability = $\frac{20 \times 15 \times 10 \times 5}{\binom{20}{4}}$.

Now let's compute that: The probability that all 4 people live on different floors is approximately 3.10%.

- (b) Now calculate the correct probability.

Solution:

- (a) ChatGPT is inconsistent about whether it is considering ordered or unordered collections. In the final probability, the numerator is an ordered sample but the denominator is an unordered sample.

- (b) There are two ways to solve this problem: using ordered or unordered sampling (as long as we are **consistent**). Note that it doesn't matter whether we sample persons or apartments as there is a one-to-one correspondence between the apartments and the persons.

If we consider the four persons/apartments as an ordered sample, the first person has 20 possibilities, the second 15, the third 10 and the fourth 5. In total there are $20 \cdot 15 \cdot 10 \cdot 5$ ordered samples giving a probability of $\frac{20 \cdot 15 \cdot 10 \cdot 5}{20 \cdot 19 \cdot 18 \cdot 17} = \frac{125}{969}$.

In we are using unordered samples, there are $\binom{20}{4}$ ways to select 4 apartments from 20 apartments. There are $5 \cdot 5 \cdot 5 \cdot 5 = 5^4$ ways to select one apartment from each of the four floors. Hence, the probability is $\frac{5^4}{\binom{20}{4}} = \frac{125}{969}$.

3. Suppose that the probability that you go to the MATH120 lecture by bike is 0.8 and the probability that you take your car is 0.2. The probability that you are late for the lecture when you ride your bike is 0.05 and the probability that you are late when you drive your car is 0.25.

- (a) What is the probability that you are late for the lecture?
 (b) If you are late to the lecture, what is the probability that you rode your bike?

Solution:

- (a)

$$\begin{aligned} P(\text{late}) &= P(\text{late}|\text{car}) \cdot P(\text{car}) + P(\text{late}|\text{bike}) \cdot P(\text{bike}) \\ &= 0.05 \cdot 0.8 + 0.25 \cdot 0.2 \\ &= 0.09 \end{aligned}$$

- (b)

$$\begin{aligned} P(\text{bike}|\text{late}) &= \frac{P(\text{late}|\text{bike}) \cdot P(\text{bike})}{P(\text{late})} \\ &= \frac{0.04}{0.09} = 4/9. \end{aligned}$$